



Chapter 4

Introduction to Cascading Style Sheets (CSS): Part 1

Internet & World Wide Web
How to Program, 5/e

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OBJECTIVES

In this chapter you'll:

- Control a website's appearance with style sheets.
- Use a style sheet to give all the pages of a website the same look and feel.
- Use the `class` attribute to apply styles.
- Specify the precise font, size, color and other properties of displayed text.
- Specify element backgrounds and colors.
- Understand the box model and how to control margins, borders and padding.
- Use style sheets to separate presentation from content.

4.1 Introduction

▶ Cascading Style Sheets 3 (CSS3) 階層式樣式表

- To specify the **presentation** (呈現方式) of elements separately from the **content** (網頁內容) of the **document**.

(Ex: 新聞、Blog)

- 註：網頁內容與樣式獨立設計，便可輕易套用不同樣式

▶ CSS validator 線上驗證CSS



網址：<http://jigsaw.w3.org/css-validator/>

- This can help you make sure that **your code is correct** (程式正確) and work well on **CSS3-compliant browsers** (瀏覽器相容).

- 註：透過線上驗證，可確保程式正確運作在CSS相容之瀏覽器

4.2 Inline Styles

▶ Inline style (內嵌式樣式)

- Embed the HTML5 **attribute style** into an element's.

- 註：透甲內嵌**style**屬性方式在某標籤中

<... **style**="屬性:數值; 屬性:數值..."> (屬性, ex: 寬度、高度..)

- ▶ Each CSS **attribute** (屬性) is followed by a **colon** ":" (冒號) and the **value of the attribute** (數值).

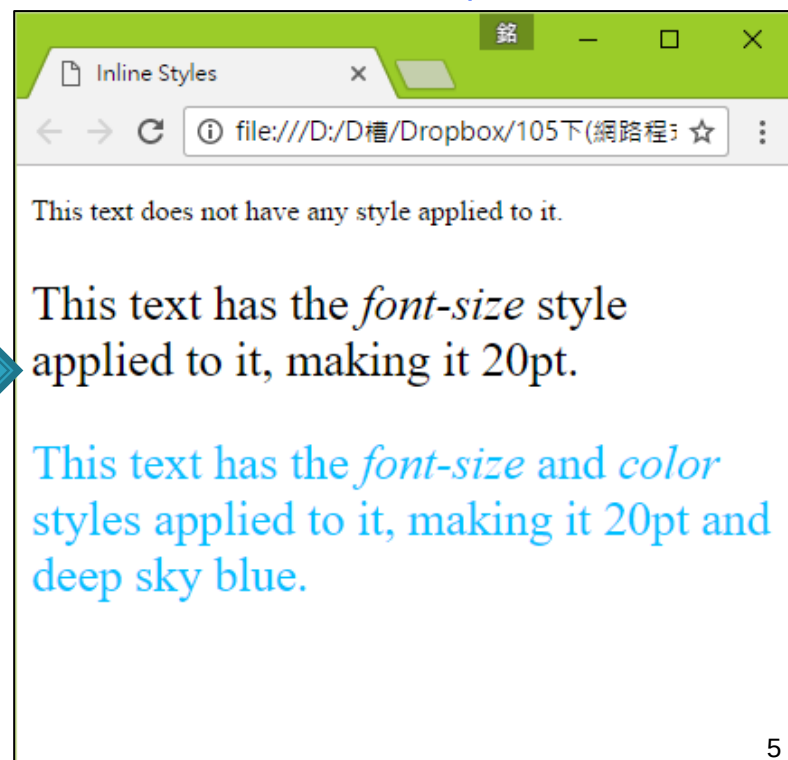
- Multiple **attributes** (多項屬性) are separated by a **semicolon** ";" (分號) (三段文字、大小不同)

```
inline.html x
1 <!DOCTYPE html>
2
3 <!-- Fig. 4.1: inline.html -->
4 <!-- Using inline styles -->
5 <html>
6 <head>
7   <meta charset = "utf-8">
8   <title>Inline Styles</title>
9 </head>
10 <body>
11   <p>This text does not have any style applied to it.</p>
12   <!-- The style attribute allows you to declare -->
13   <!-- inline styles. You should separate multiple -->
14   <!-- style properties with a semicolon. -->
15   <p style = "font-size: 20pt;">This text has the
16     <em>font-size</em> style applied to it, making it 20pt.
17   </p>
18   <!-- style embedded in p tag, attribute font-size: value 20pt, color "deep sky blue" -->
19   <p style = "font-size: 20pt; color: deepskyblue;">
20     This text has the <em>font-size</em> and
21     <em>color</em> styles applied to it, making it
22     20pt and deep sky blue.</p>
23 </body>
24 </html>
```

(練習Trace code)

//style embedded in p tag, attribute font-size: value 20pt

// style embedded in p tag, attribute font-size: value 20pt, color "deep sky blue"



4.2 Inline Styles (Cont.)

- ▶ **color**: 數值或顏色 // 顏色屬性
 - To set **text color** (文字顏色) and can use **color names** (顏色名稱) or **hexadecimal codes** (顏色的16進位碼) as its **value**.
 - 註：顏色屬性用於設定文字色彩，可用顏色名稱或顏色代碼(16進位)。
 - Appendix B shows more **hexadecimal color codes** and **color names**
- Extended colors: <http://www.w3.org/TR/css3-color/>
 - 註：更多的顏色代碼可參考附錄B或以上網址



Color name	Value	Color name	Value
aqua //淺綠色	#00FFFF	navy //海軍藍(藏青色)	#000080
black	#000000	olive //橄欖綠	#808000
blue	#0000FF	purple	#800080
fuchsia //紫紅色	#FF00FF	red	#FF0000
gray	#808080	silver	#C0C0C0
green	#008000	teal //靛綠(藍綠色)	#008080
lime //酸橙色(青橙綠色)	#00FF00	yellow	#FFFF00
maroon //栗色(褐紫紅)	#800000	white	#FFFFFF

Lab: 使用3種不同的顏色看看(名稱/16進位碼)

Fig. 4.2 | HTML standard colors and hexadecimal RGB values

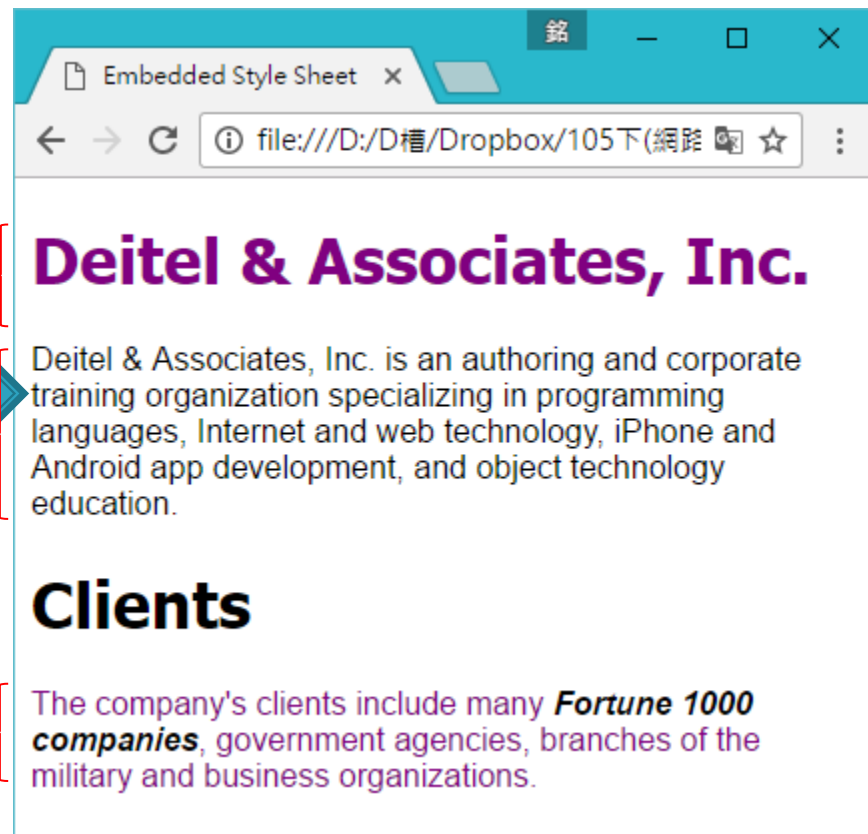
4.3 Embedded Style Sheets

(另一種方式)

Embedded style sheets 嵌入樣式表

- ▶ To embed a CSS3 document (CSS文檔) in an HTML5 document's head section.
 - 註：將樣式表寫定獨立標籤，再用嵌入方式使用

```
embedded.html
1 <!DOCTYPE html>
2
3 <!-- Fig. 4.3: embedded.html -->
4 <!-- Embedded style sheets. -->
5
6 <html>
7   <head>
8     <meta charset = "utf-8">
9     <title>Embedded Style Sheet</title>
10
11     <!-- this begins the style sheet section --> //定義4種style
12     <style type = "text/css">
13       em { font-weight: bold; // (1) 斜體em: 粗體+黑色
14         color: black; } // (2) h1: 指定"預設"字型
15       h1 { font-family: tahoma, helvetica, sans-serif; }
16       p { font-size: 12pt; // (3) p: 指定字體大小12及預設字型
17         font-family: arial, sans-serif; }
18       .special { color: purple; } // (4) 新定義樣式"special", 顏色為紫色
19     </style>
20   </head>
21   <body>
22     <!-- this attribute applies the .special style class -->
23     <h1 class = "special">Deitel & Associates, Inc.</h1>
24
25     //指定樣式: special
26     <p>Deitel & Associates, Inc. is an authoring and
27     corporate training organization specializing in
28     programming languages, Internet and web technology,
29     iPhone and Android app development, and object
30     technology education.</p>
31
32     //套用p樣式: 字體大小12及預設字型
33     <h1>Clients</h1>
34     <p class = "special">The company's clients include many
35     <em>Fortune 1000 companies</em>, government agencies,
36     branches of the military and business organizations.</p>
37
38     //套用em樣式: 粗體+黑色 //內層樣式, 優先套用
39   </body>
40 </html>
```



4.3 Embedded Style Sheets (cont.)

<style type = "text/css">...</style> *style標籤及型態*

- ▶ Style's **type attribute** specifies the **MIME type** by "**text/css**".
 - 註：告知瀏覽器是**CSS描述**
- ▶ Some **MIME types** used in this book.

MIME type	Description
text/css //用於CSS描述	CSS documents
image/png //用於png描述	PNG images
text/javascript //用於JavaScript描述	JavaScript markup
text/plain //用於純文字描述	Plain text
image/jpeg //用於jpeg描述	JPEG image
text/html //用於html描述	HTML markup

Fig. 4.4 | A few common MIME types.

4.3 Embedded Style Sheets (cont.)

▶ The style sheet 樣式表

- Describe the **CSS rules (樣式規則)** and uses a **CSS selector (CSS篩選器)** to specify the elements to be styled according to a rule.
 - 註：使用**CSS篩選器**針對**指定標籤**來**套用樣式規則**。
- Each style rule body is enclosed in curly braces (大括弧) “{” and “}”.**
 - 註：樣式規則的內容，使用**大括弧**包起來。

指定標籤 →

```
<style type = "text/css">
em      { font-weight: bold;
          color: black; }
h1      { font-family: tahoma, helvetica, sans-serif; }
p       { font-size: 12pt;
          font-family: arial, sans-serif; }
.special { color: purple; }
</style>
```

▶ **em element:** 斜體標籤

▶ **font-weight:** specifies the “boldness” (粗體) of text.

- Possible values are:
 - bold (粗體)
 - normal (預設)
 - bolder (比粗體還粗)
 - lighter (比預設還細)
 - 註：Boldness 還可用**數字**描述，以**100的倍數** (e.g., 100, 200, ..., 900).
 - 註：預設 (normal) 為 **400**, 粗體 (bold) 為 **700**.

4.3 Embedded Style Sheets (Cont.)

.樣式名稱 {...} //定義新樣式類型

- ▶ **Style Classes (樣式類型)** declarations are started by a period “.”

- ▶ 註：用 . 開始定義新樣式

- ▶ The defined styles can be applied to *any element*.

- ▶ 註：可套用於任何標籤

- ▶ For example:

```
<style type = "text/css">
  em      { font-weight: bold;
            color: black; }
  h1      { font-family: tahoma, helvetica, sans-serif; }
  p       { font-size: 12pt;
            font-family: arial, sans-serif; }
  .special { color: purple; }
</style>
```

```
<body>
  <!-- this attribute applies the .special style class -->
  <h1 class = "special">Deitel & Associates, Inc.</h1>

  <p>Deitel & Associates, Inc. is an authoring and
    corporate training organization specializing in
    programming languages, Internet and web technology,
    iPhone and Android app development, and object
    technology education.</p>
```

Lab: 自訂2種不同的樣式看看

- ▶ You can also declare **id selectors**.

- ▶ If **the element** has an **id**, you can declare by **#idname** to specify that element's style.

- ▶ 註：若標籤有id名稱，則可以**#idname**作為selector來作描述規則。

5.3 Embedded Style Sheets (Cont.)

```
<style type = "text/css">
  em      { font-weight: bold;
            color: black; }
  h1      { font-family: tahoma, helvetica, sans-serif; }
  p       { font-size: 12pt;
            font-family: arial, sans-serif; }
  .special { color: purple; }
</style>
```

font-family //預設字型屬性

- ▶ To specify a type of font (字型 型態) instead of a specific font.
 - 註：使用字型家族描述，避免瀏覽器不支援某字型。

Generic font families	Examples (常見字型家族，包含多種字型)
serif	times new roman, georgia
sans-serif	arial, verdana, futura
cursive	script
fantasy	critter
monospace	courier, fixedsys

字型	呈現
serif	Show serif font
sans-serif	Show sans-serif font
cursive	Show cursive font
fantasy	Show fantasy font
monospace	monospace font

Fig. 4.5 | Generic font families

4.3 Embedded Style Sheets (Cont.)

```
<style type = "text/css">
  em      { font-weight: bold;
            color: black; }
  h1      { font-family: tahoma, helvetica, sans-serif; }
  p      { font-size: 12pt;
            font-family: arial, sans-serif; }
  .special { color: purple; }
</style>
```

font-size //字型大小屬性

(相對大小-適用不同螢幕)

► To specify the size by point size (pt, 點數) or relative value, e.g.,

- xx-small (非常小), x-small (極小), small (小), smaller (較小), medium (中), large (大), larger (較大), x-large (極大) and xx-large (超級大).
- 註：較偏好用相對大小，因為無法確定使用者的螢幕呈現。

Applying a Style Class to nested elements //套用巢狀標籤

► The styles applied to the parent element (父標籤) also apply to the element's child elements (子標籤).

- 註：套用在父標籤的style，也會套用在子標籤上，具有繼承性質。

//父標籤<p>

```
<h1>Clients</h1>
<p class = "special"> The company's clients include many
  <em>Fortune 1000 companies</em>, government agencies,
  branches of the military and business organizations.</p>
</body>
```

//子標籤: 黑色+粗體

Clients

The company's clients include many **Fortune 1000 companies**, government agencies, branches of the military and business organizations.

5.4 Conflicting Styles

- ▶ Styles may be defined by a **user** (使用者 / 作業系統), an **author** (網頁開發者) or a **user agent** (瀏覽器).
 - Styles defined by **authors** take precedence over styles defined by the **user**.
 - Styles defined by the **user** take precedence over styles defined by the **user agent**.
 - 註：樣式方面，網頁開發者設定優先於使用者本機(作業系統)設定優先於瀏覽器設定。

(網頁開發者→使用者本機→瀏覽器)

- ▶ **text-decoration: //文字裝飾屬性**

underline (字底畫線)

overline (字頂畫線)

line-through (字中畫線：刪除線)

none (無) (例如：超連結，不顯示底線)

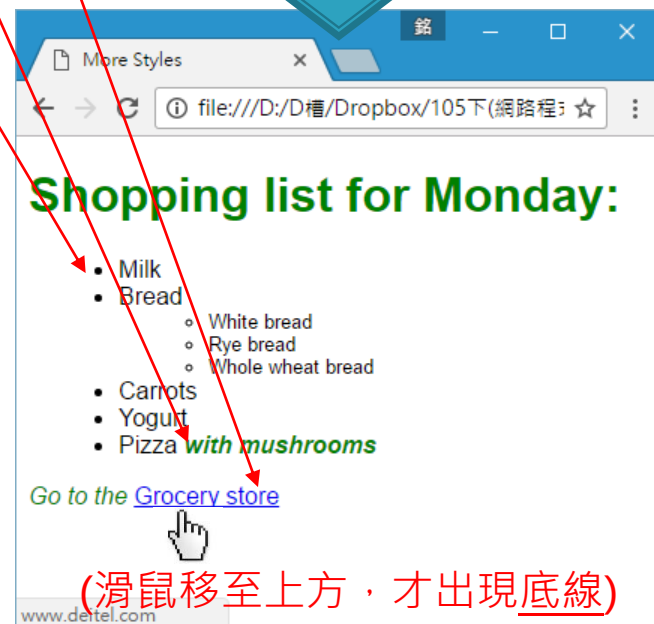
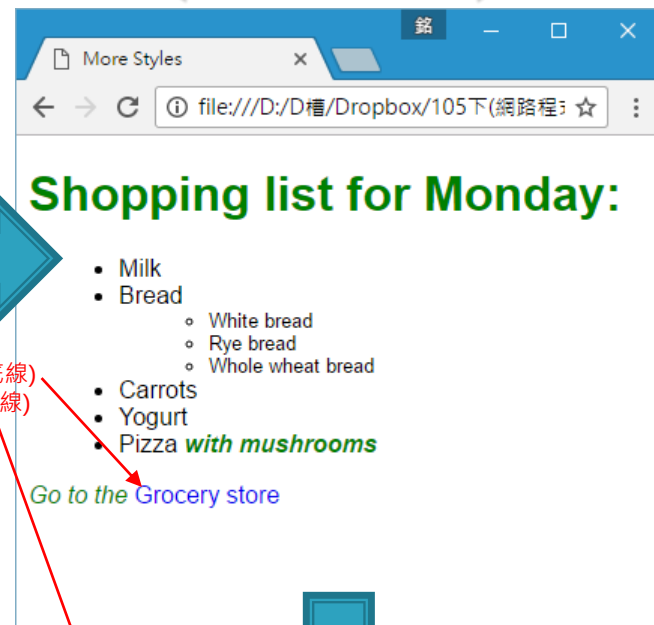
Lab: 練習3種不同的字體裝飾線屬性

5.4 Conflicting Styles (Cont.)



```
1 <!DOCTYPE html>
2
3 <!-- Fig 4.6: advanced.html -->
4 <!-- Inheritance in style sheets. -->
5 <html>
6 <head>
7   <meta charset = "utf-8">
8   <title>More Styles</title>
9   <style type = "text/css">
10     body { font-family: arial, helvetica, sans-serif; }
11     a.nodect { text-decoration: none; } //a標籤下定義「新樣式nodect」(不劃底線)
12     a:hover { text-decoration: underline; } //滑鼠移到a標籤的樣式(字體劃下線)
13     //新定義
14     li em { font-weight: bold; } //li標籤內的em標籤的樣式(粗體)
15     h1, em { color: green; } //h1標籤或em標籤的樣式(字體綠色)
16     //逗號(多項)
17     ul { margin-left: 20px; } //ul標籤樣式(左側空白20點)
18     ul ul { font-size: .8em; } //ul標籤下的ul標籤樣式(字體縮小)
19     p { color: red; } //p標籤(字體紅色)
20   </style>
21 </head>
22 <body>
23   <h1>Shopping list for Monday:</h1>
24   <ul>
25     <li>Milk</li>
26     <li>Bread // ul標籤下的ul標籤樣式(字體縮小)
27       <ul>
28         <li>White bread</li>
29         <li>Rye bread</li>
30         <li>Whole wheat bread</li>
31       </ul>
32     <li>Carrots</li>
33     <li>Yogurt</li>
34     <li>Pizza <em>with mushrooms</em></li>
35   </ul>
36   //子標籤em有較高優先權於父標籤p(字體綠色)
37   <p><em>Go to the</em> //子標籤em繼承父標籤p的樣式(字型大小)
38     <a class = "nodect" href = "http://www.deitel.com">
39       Grocery store</a>
40   </p>
41 </body>
```

(輸出)



4.4 Conflicting Styles (Cont.)

► Pseudoclasses (偽類別，某些標籤內建的類別)

- **a: hover** is a pseudoclass of a element and activated when **mouse cursor** is **over** an element.

- 註：偽類別為某些標籤內建的類別，比如 **a: hover {...}** 是滑鼠移到超連結上面的樣式

字體大小單位 (相對長度)

- **px** (pixels, 像素量, ex: **12px**，取決於螢幕解析度)
- **em** (字體大寫的高度, ex: **1.2em**)
- **ex** (字體小寫的高度, 較少用)
- **%** (預設字體大小的百分比, ex: **77%**)

Lab: 練習3種不同的字體大小單位
(相對單位)

字體大小單位 (絕對長度)

- **in** (inches, 英吋, ex: **2in**)
- **cm** (centimeters, 公分, ex: **3cm**)
- **mm** (millimeters, 公厘, ex: **1mm**)
- **pt** (points; 1 pt = 1/72 in, 點數, ex: **2pt**)
- **pc** (picas; 1 pc = 12 pt, 派卡, ex: **1pc**, 較少用)

Lab: 練習3種不同的字體大小單位
(絕對單位)

4.5 Linking External Style Sheets

- ▶ **External style sheets** //獨立的樣式表單
 - Separate documents that contain only **CSS rules**.
 - 註：獨立成一個樣式檔案(.css)，只包含CSS規則。
//概念: 類似c程式include某些函式庫
- ▶ Have a **uniform look** (一致呈現) for a website
 - 註：網站分頁可“共用”相同樣式 (易於更換及管理)
 - 註：這種概念也被視為換膚(skinning).
- ▶ **CSS comments** CSS文字註解
 - 註：可放置於標籤內(<...style="..." ...>), 嵌入樣式(<style>...</style>)及外部樣式(external style sheets)
 - 註：類似C語言註解：開始用 /*，結束用 */.

4.5 Linking External Style Sheets (Cont.)

(.css style檔案)

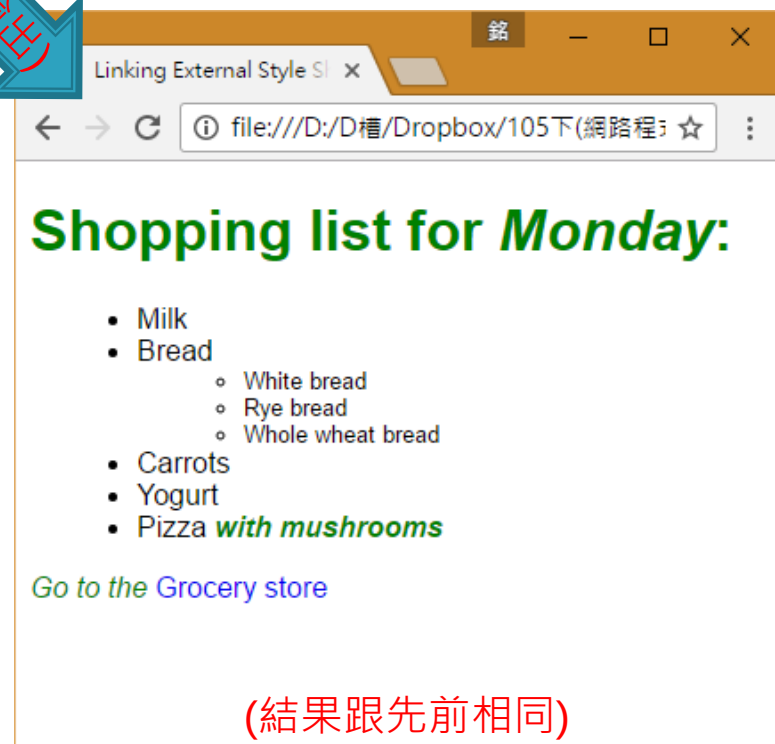
```
external.html
1 <!DOCTYPE html>
2
3 <!-- Fig. 4.8: external.html -->
4 <!-- Linking an external style sheet. -->
5 <html>
6 <head>
7   <meta charset="utf-8">
8   <title>Linking External Style Sheets</title>
9   <link rel = "stylesheet" type = "text/css"
10     href = "styles.css">
11 </head>
12 <body>
13   <h1>Shopping list for <em>Monday</em>:</h1>
14
15   <ul>
16     <li>Milk</li>
17     <li>Bread
18       <ul>
19         <li>White bread</li>
20         <li>Rye bread</li>
21         <li>Whole wheat bread</li>
22       </ul>
23     </li>
24     <li>Carrots</li>
25     <li>Yogurt</li>
26     <li>Pizza <em>with mushrooms</em></li>
27   </ul>
28
29   <p><em>Go to the</em>
30     <a class = "nodec" href = "http://www.deitel.com">
31       Grocery store</a>
32   </p>
33 </body>
34 </html>
```

//引用外部css
單標籤<link>

//描述外部資訊

//其他不變

```
external.html styles.css
1 /* Fig. 4.7: styles.css */
2 /* External style sheet */
3 body { font-family: arial, helvetica, sans-serif; }
4 a.nodec { text-decoration: none; }
5 a:hover { text-decoration: underline; }
6 li em { font-weight: bold; }
7 h1, em { color: green; }
8 ul { margin-left: 20px; }
9 ul ul { font-size: .8em; }
10 p { color: red; }
```



(結果跟先前相同)

4.5 Linking External Style Sheets (Cont.)

```
<head>
  <meta charset="utf-8">
  <title>Linking External Style Sheets</title>
  <link rel = "stylesheet" type = "text/css"
        href = "styles.css">
</head>
```

<link rel="..." type="..." href="..."> //link單標籤

▶ **rel = "stylesheet"**

- declares the **linked document** to be a **stylesheet** for the document
- 註：rel = "stylesheet"描述：被連結的檔案是一個樣式表

▶ **type = "text/css"**

- specifies the MIME type of the related document
- 註：告知瀏覽器是text/css型態文件

▶ **href = "styles.css"**

- provides the URL for the document containing the style sheet
- 註：提供樣式表的網址來源

4.6 Positioning Elements: Absolute Positioning, z-index

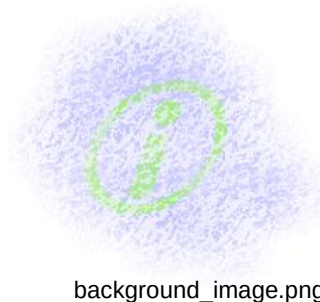
- ▶ **CSS position property (位置屬性)**
 - Describe **absolute positioning (描述絕對位置)**, which provides greater control over where on a **page elements** reside
 - 註：用於**標籤內**，描述**標籤的絕對位置**
 - Positions it according to **distance** from the **top (上邊界)**, **left (左邊界)**, **right (右邊界)** or **bottom margin (下邊界)** of its parent element
 - 註：**定位距離**根據其**父標籤**的**上邊界、左邊界、右邊界、下邊界 (之距離)**

4.6 Positioning Elements: Absolute

Positioning, z-index



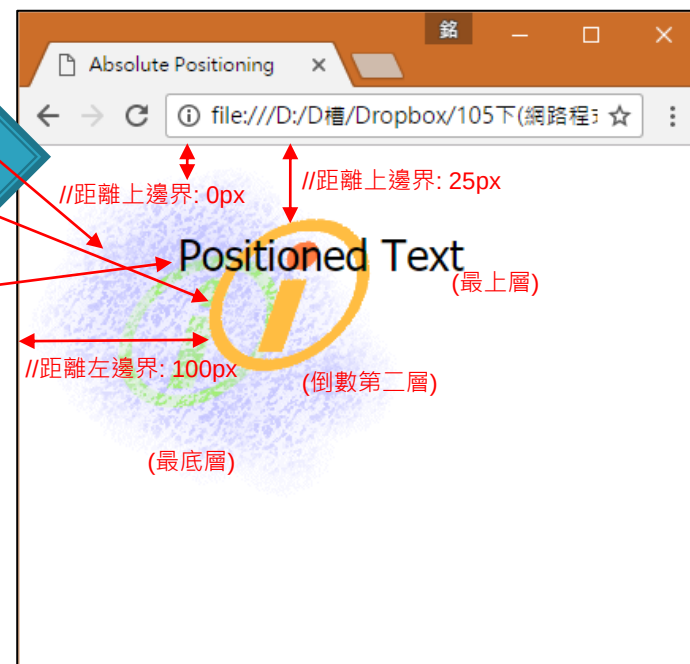
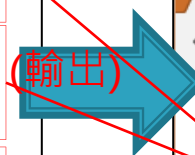
```
positioning.html x
1 <!DOCTYPE html>
2
3 <!-- Fig 4.9: positioning.html -->
4 <!-- Absolute positioning of elements. -->
5 <html>
6 <head>
7 <meta charset = "utf-8">
8 <title>Absolute Positioning</title>
9 <style type = "text/css">
10 .background_image { position: absolute; //描述背景圖絕對位置
11 //定義 新類別 top: 0px; //距離上邊界: 0px
12 left: 0px; //距離左邊界: 0px
13 z-index: 1; } //圖層: 1 (最下層)
14 .foreground_image { position: absolute; //描述前景圖絕對位置
15 //定義 新類別 top: 25px; //距離上邊界: 25px
16 left: 100px; //距離左邊界: 100px
17 z-index: 2; } //圖層: 2 (倒數第2層)
18 .text { position: absolute; //描述文字 絕對位置
19 //定義 新類別 top: 25px; //距離上邊界: 25px
20 left: 100px; //距離左邊界: 100px
21 z-index: 3; //圖層: 3 (倒數第3層)
22 font-size: 20pt; //字體: 20pt //預設字體家族
23 font-family: tahoma, geneva, sans-serif; }
24 </style>
25 </head>
26 <body>
27 <p><img src = "background_image.png" class = "background_image"
28 alt = "First positioned image" /></p>
29
30 <p><img src = "foreground_image.png" class = "foreground_image"
31 alt = "Second positioned image" /></p>
32
33 <p class = "text">Positioned Text</p>
34 </body>
35 </html>
```



background_image.png



foreground_image.png



4.6 Positioning Elements: Absolute Positioning, z-index (Cont.)

z-index: 數值 //圖層數值屬性

- ▶ To allow overlapping elements
- ▶ Elements that have **higher z-index** values are displayed **in front of elements** with lower z-index values
 - 註：允許將元素重疊，**z-index**值越大，在越“上層”。

display: //呈現方式屬性

- ▶ allows a programmer to decide if an element is displayed as a
 - **block element** (區塊形式呈現)
 - **inline element** (不換行呈現)
 - **none** (不呈現/隱藏)

text-align: //文字呈現: 置中/置左/置右

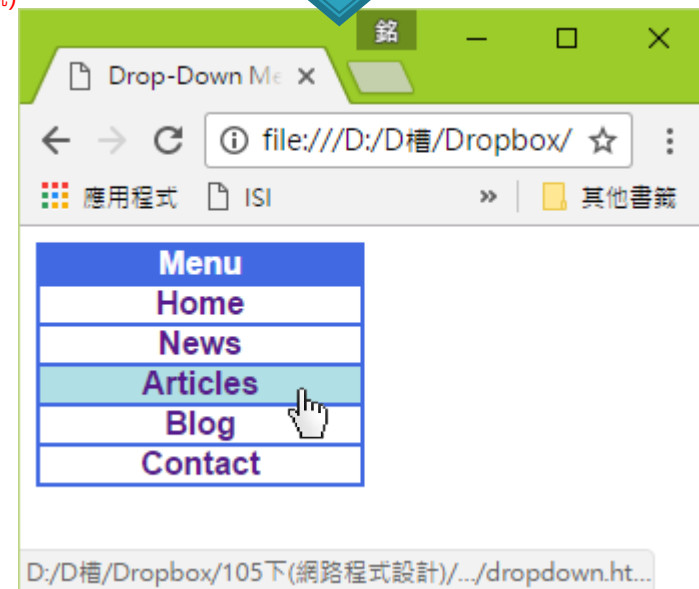
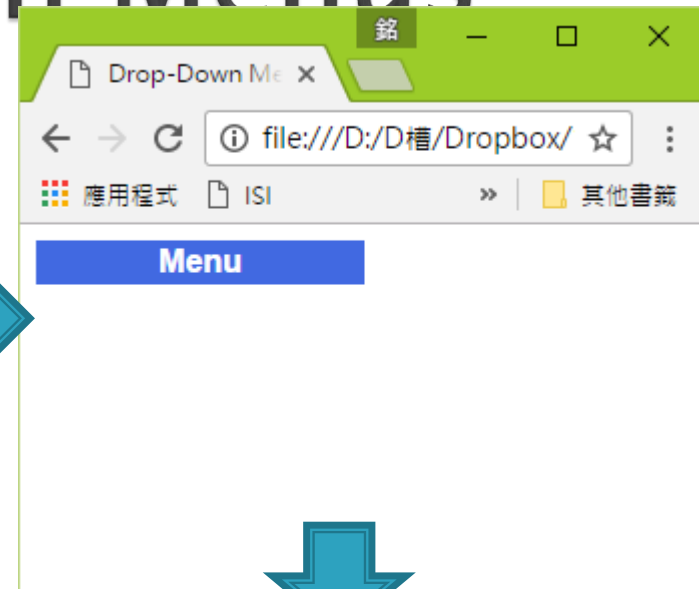
- ▶ Most elements are **left-aligned** (靠左) by **default** (預設)
 - 註：大部分元素都是預設靠左
- ▶ The values for the **text-align** property include
 - center (置中),
 - left (置左),
 - right (置右).

4.12 Drop-Down Menus

```
dropdown.html x
1 <!DOCTYPE html>
2
3 <!-- Fig. 4.17: dropdown.html -->
4 <!-- CSS drop-down menu. -->
5 <html>
6 <head>
7 <meta charset = "utf-8">
8 <title>
9   Drop-Down Menu
10 </title>
11 <style type = "text/css">
12   body
13     { font-family: arial, sans-serif }
14   nav
15     { font-weight: bold; //描述字體: 粗體
16       color: white; //描述顏色: 白色
17       border: 2px solid RoyalBlue;
18       text-align: center; //文字呈現: 置中
19       width: 10em; //導覽列寬度: 10em
20       background-color: RoyalBlue; } //背景顏色: 深藍
21   nav ul
22     { display: none; //呈現方式: 隱藏
23       list-style: none; //編號方式: 無 (ex: 數字/羅馬符號)
24       margin: 0; //邊界寬: 0
25       padding: 0; //邊界距離: 0
26   nav:hover ul
27     { display: block; } //呈現方式: 區塊出現
28   nav ul li
29     { border-top: 2px solid RoyalBlue; //上邊界: 2px
30       background-color: white; //元件背景色: 白
31       width: 10em; //元件寬度: 10em (相同)
32       color: black; //元件顏色: 黑
33   nav ul li:hover
34     { background-color: PowderBlue; } //子項顏色: 淺藍
35   a // 超連結
36     { text-decoration: none; } //文字底線: 無
37 </style>
38 </head>
39 <body>
40 <nav>Menu
41 <ul>
42 <li><a href = "#">Home</a></li>
43 <li><a href = "#">News</a></li>
44 <li><a href = "#">Articles</a></li>
45 <li><a href = "#">Blog</a></li>
46 <li><a href = "#">Contact</a></li>
47 </ul>
48 </nav>
49 </body>
50 </html>
```

(從範例學習)

(輸出)



4.6 Positioning Elements: Absolute Positioning, z-index (Cont.)

width, height //寬度及高度屬性

- ▶ To set the **width** (寬度) and **height** (高度) of an element on screen.
- ▶ The **width** and **height** values also can be specified as **relative** (相對長) or **absolute lengths** (絕對長).
 - 註：元件的寬度及高度可以用相對單位(ex: %)及絕對單位(ex: px)描述。

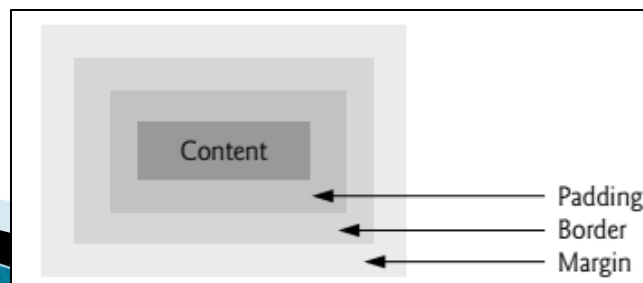
background-color //背景顏色

- ▶ To fill color in areas of the element
 - 註：設定元件的背景顏色 (ex: background-color: white;)

Menu
Home
News
Articles
Blog
Contact

margin, padding //外框屬性

- ▶ **margin**: to set the **space** between the **outside** of an element's.
 - 註：**margin**描述元件的外圍邊界大小
 - 註：元件的四邊界可以各自設定: **margin-top** (上邊界), **margin-right** (右邊界), **margin-left** (左邊界), **margin-bottom** (下邊界).
- ▶ **padding**: to determine the **distance** between the content inside an element.
 - 註：**padding**描述元件的外圍距離長度
 - 註：元件的四週外圍距離可以各自設定: **padding-top** (上邊距離), **padding-right** (右邊距離), **padding-left** (左邊距離), **padding-bottom** (下邊距離).

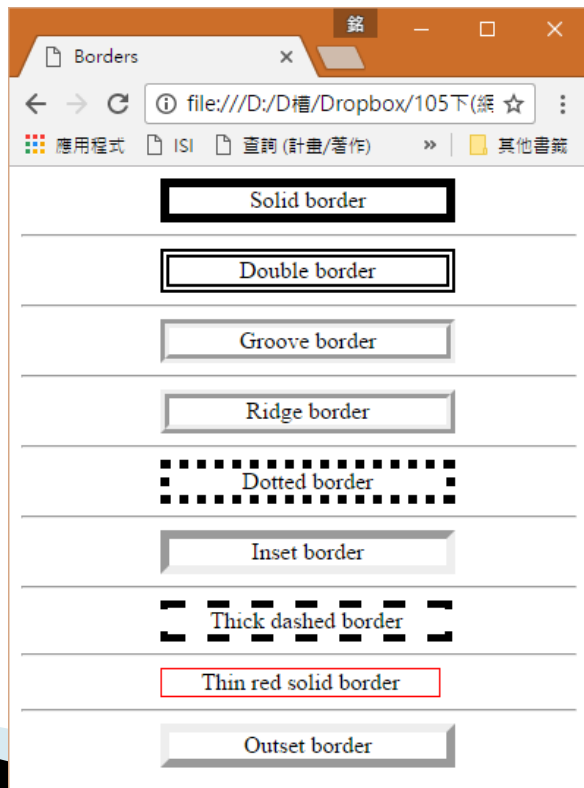


(使用看看，能更了解)

4.6 Positioning Elements: Absolute Positioning, z-index (Cont.)

border-width, border-color, border-style // 邊界寬度, 顏色, 樣式

- ▶ **border-width** (邊界寬): 可指定寬度 (如 : em, ex, px) 或 特定詞: **thin** (細), **medium** (中等) or **thick** (粗).
- ▶ **border-color** (邊界顏色): 可設置任何顏色.
- ▶ **border-style** (邊界樣式): 可指定樣式, 如 : none (不顯示), hidden (隱藏), dotted (虛線), dashed (長線段), solid (粗實線), double (雙線), groove (凹槽線), ridge (凸槽線), inset (內嵌線) and outset (外嵌線).



//粗實線

//雙線

//凹槽線

//凸槽線

//虛線

//內嵌線

⋮

//外嵌線



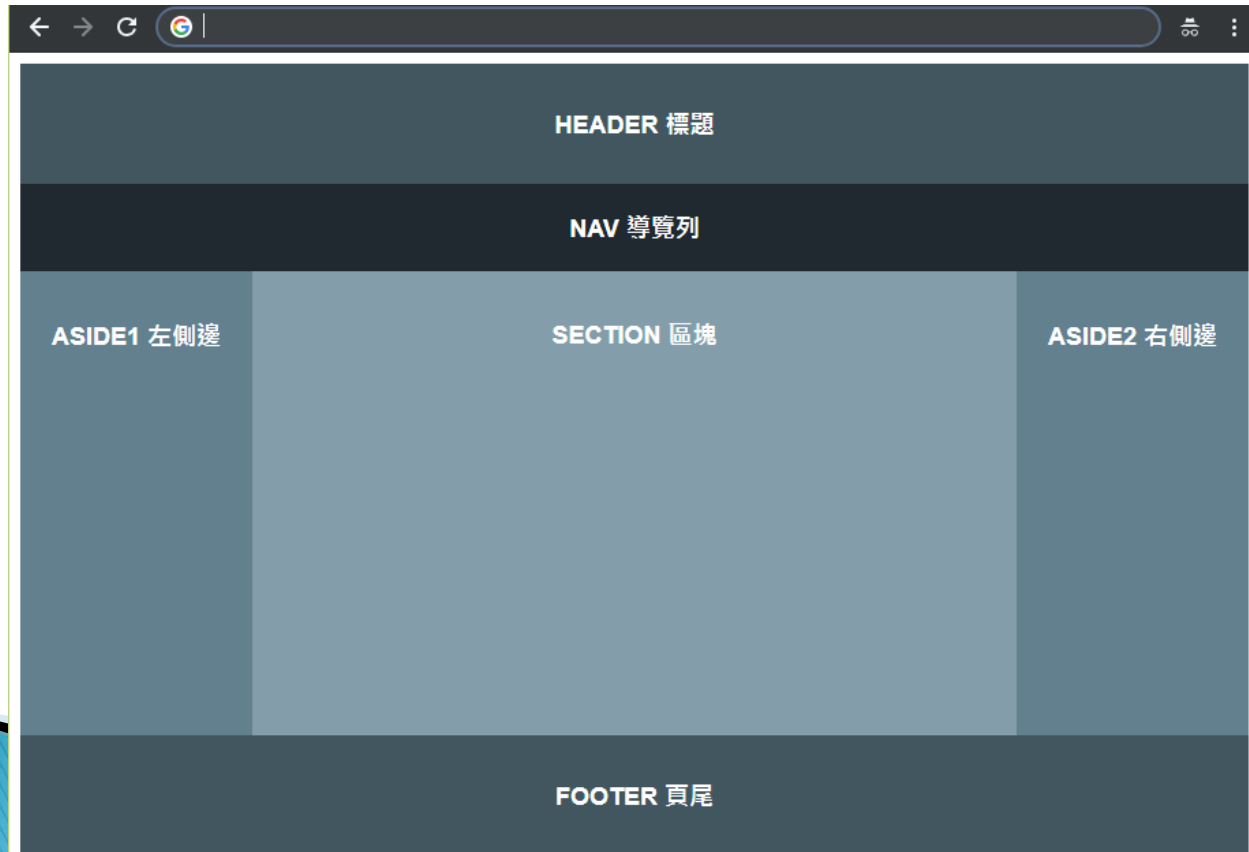
Lab

(上機練習)



In-Class Exercise #1

- ▶ 課中練習：請利用 **CSS** 結合上週 **Lab03** 框架標籤 (header, nav, section, aside, footer等)，構成如下版面：

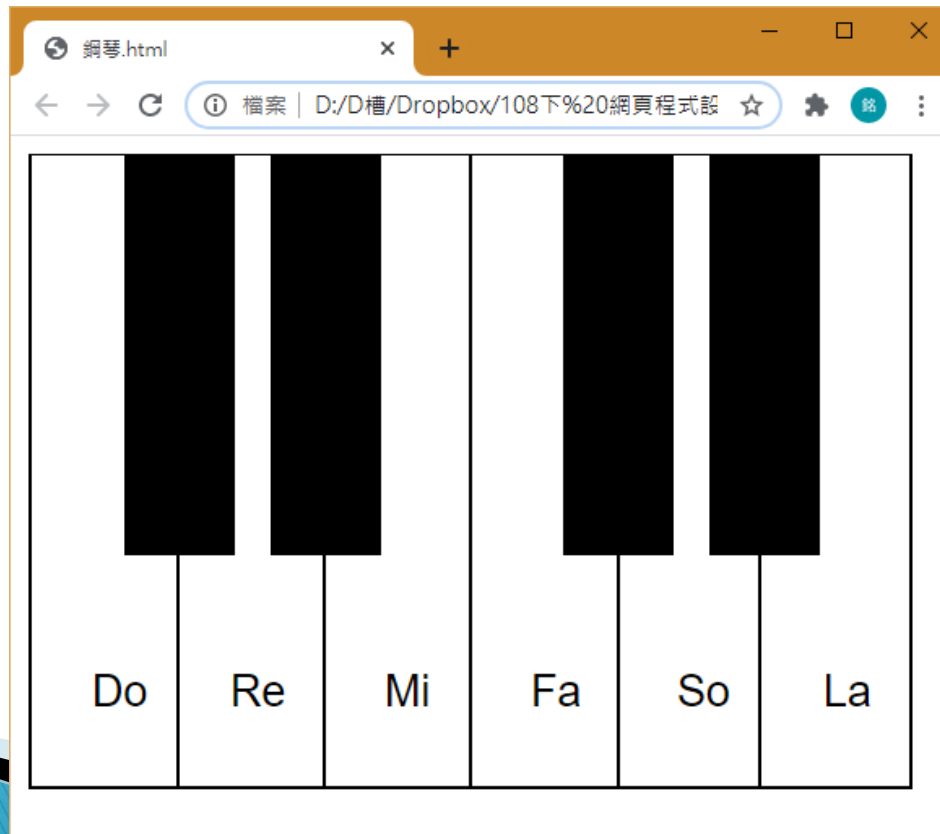


提示:

1. header 週圍若有空隙
可使用: `padding: 20px;` 填補
2. 兩個aside:
一個使用 `float: right;`
另一使用 `float: left;`
(意涵：區塊置右及置左)

In-Class Exercise #2

- ▶ 課中練習：透過圖層方式，將2張圖片(白鍵 white_key.png)、(黑鍵 black_key.png)及音階文字，建構出下方鋼琴鍵：
 - 圖片來源：FTP\課程講義 (Course slides)\Ch04課程檔案\課程範例\In-Class exercise 2\





(Bonus #1)

- 加分題 (*稍難*)：嘗試建構出多階層式選單(不限於課堂教的方式)，如下：



注意：

- 本週五 23:59 前上傳到ftp (否則算遲交)：
<ftp://111webdesign:webdesign@120.114.190.22:4000>
- Lab04
- 請以"學號"建立資料匣 (否則扣分)